

The objective of this assignment is for you to play with reinforcement learning. We will use the environments in OpenAI Gym, the maintained fork of which is in Gymnasium (<https://gymnasium.farama.org/>). The programming language used is python.

Requirements:

- Go to the website above and setup the environments following the instructions there.
- Choose at least two tasks for your project. At least one task should be from the Atari environment. You can choose the other from any of the other environments.
- There is no limitation/requirement on the type of RL algorithms you can use.
- You can probably find lots of web resources and code deposits for the tasks here. You can certainly consult them and even reuse some code. This will help you get started well, as RL is known to be more difficult to train than, say, simple classifiers. You should provide references to the resources you use though.
- You are responsible for designing and executing the experiments on different aspects that might affect the performance of the agents.
- The difficulty levels of your selected tasks are not that important. Some of the tasks are very simple, like inverted pendulum or cliff walking. However, you can still get good grades if you can do meaningful and interesting experiments on them.

Submission:

- A report in PDF format (maximum 10 pages single-spaced). It should contain the following sections:
 - A brief statement of your research question and motivation in plain text.
 - Description of the methods used. Provide references for public libraries, open-source codes, and/or AI tools used.
 - Description of your experiments, include evaluation results and examples.
 - A "discussion" section as described above.
 - List of references.
 - Include your program code as an appendix (not counting toward the 10-page limit), starting from a separate page. The code listing should be well-organized and contain comments that help the readers understand your code.
- Additional notes about the report:
 - Submit your report through E3. Late submissions are accepted for up to 5 days, with a deduction of 10% per day.
 - Make sure your student ID is within the filename as well as on the first page.
 - For the description of methods: You should expect the reader to read this part of your report without the code listing. Do NOT write your report like a documentation of your program.
 - Experimental results should come with sufficient description/summarization and discussions. When you list your results, always ask yourself this question: "What message do I want my reader to learn from these numbers?" And then put your answer to text.
 - Presentation of results: Use tables formatted in your report, NOT screenshots. Use charts or plots when appropriate.
 - Use at least size-12 fonts for your main text and at least size-10 fonts in tables and figures.